

EE/CprE/SE 491 — sdmay27-37

Corn Harvest Data Collection Application

Week 02 Report

09/22/2026 – 09/29/2026

Client:	Bayer (Jason Knoche)
Faculty Advisor:	Dr. Phillip Jones

Team Members

Team Member	Role
Connor Moroney	Component Design Lead
Jonah Samuels	Client Interaction Lead
Carter Voelkel	Team Organization Lead
Megan Chng	Testing Lead
Cael Shroff	Documentation & Reporting Lead
Owen Gilbertson	Research Lead

Weekly Summary

Project Direction

Now that we have a high-level understanding of the project, this week we focused on the specifics. The biggest decision was the platform. Data collectors will be working in the field, often walking rows with one hand free, so we went with a mobile design they can pull up on their phones without any setup. We also decided to build a web application instead of a native one. That means nothing to install, no app store submission or review process, and no need to pre-load devices for each user. Updates are easier too, since every collector gets the latest version as soon as they open the page.

Benefits for Researchers

Building for the web also helps a second group of users. Collectors will use the app on their phones in the field, but researchers can open the same application on a computer and see the collected data in a more digestible format. A bigger screen makes it easier to look at readings across many rows at once, compare results, and notice patterns that would be hard to catch on a phone. Since everything lives in one web app, collectors and researchers work from the same data without exporting or transferring anything between tools.

Connectivity Challenge

The web approach does come with one big hurdle, which came up in our bi-weekly meeting with Jason. Corn fields have unreliable cell reception, and a web app that needs a constant connection would fail right where it's needed most. We still have to figure out how the app can keep working offline. Some options are caching the app itself, storing readings on the device, and syncing them to the server once the collector gets a signal again.

Prototype

To test our ideas, we built a prototype that simulates the core data collection process. A collector manually enters moisture percentage readings for a row, and the app requires several readings per row instead of just one. It averages them to get a more reliable value and also checks how spread out the readings are. If the variance is too high, which could mean a bad measurement or a typing mistake, the collector has to re-enter the data. This puts a basic quality check into the collection process itself so it doesn't have to be cleaned up later.

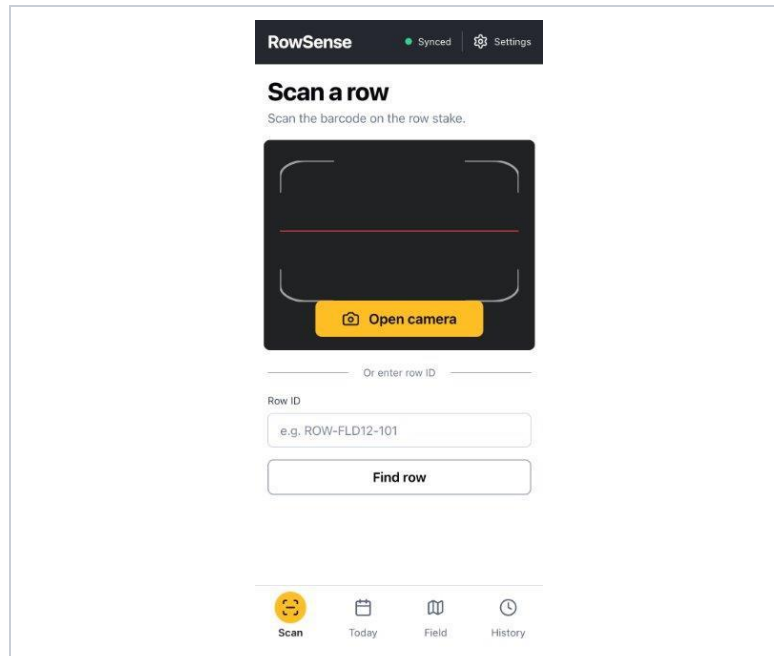


Figure 1. Row scan screen from the RowSense prototype.

QR Code Row Tags

In the same meeting, we were shown an example of a tag that identifies each row with a QR code. This opens up a useful possibility. Instead of picking or typing in a row number, a collector could scan the tag with their phone camera and go straight to that row's entry screen. That would cut down on mistakes from entering data under the wrong row and would speed up collection.



Figure 2. Example row tag shown by the client, with QR codes for row identification.

Client Meeting

Our bi-weekly meeting with Jason Knoche, our industry client, was the second time we've met with him. We had sent him some questions beforehand and came in with answers to them, which let us present the prototype with a clearer picture of what he needs. Seeing a working example gave him something concrete to react to. His feedback sharpened our understanding of the requirements and helped us figure out where the project goes from here.

Past Week Accomplishments

Carter Voelkel

Prepped lightning talk slides to finish up with the team for Tuesday's presentation. Setup GitHub for access to the project. Also researching HTML5 and other options to store data on a web app while waiting to get a better connection to the internet.

Jonah Samuels

Built a prototype UI based on our first meeting with the Client. Presented prototype for feedback in 2nd meeting with Client. Brainstormed list of questions about the current data collection process. Practiced lightning-talk presentation during weekly group meeting.

Connor Moroney

Defined three user roles for the application (Collector, Researcher, and Supervisor) and documented what each can view and do:

Collector	Intended for field data collectors. Each Collector is assigned to a specific field. Collectors can view current and historical moisture data for their assigned field, upload new readings to it, and delete readings they uploaded themselves (for example, to correct a mistake).
Researcher	Intended for researchers who analyze moisture data. Researchers can view data across all fields but cannot upload, edit, or delete any readings.
Supervisor	Intended for the collectors' supervisors. Supervisors can assign roles to new employees, create and modify fields, and assign users to them. They can also view, upload, edit, and delete moisture data across all fields.

Getting the roles defined this early matters because they shape how the rest of the app gets built. Every screen, database query, and sync rule depends on knowing who is allowed to do what, and it is a lot harder to add that later than to plan for it now. Roles also protect the data. Collectors only reach their assigned field and can only delete their own readings, so one wrong tap can't wipe out someone else's work. Researchers are read-only, which keeps the data they analyze from being changed by accident. Only Supervisors can change fields, users, and roles, so that kind of access stays with a small number of people.

Security is worth planning for now since this is the client's data. Users will need to log in, and permissions should be enforced on the server, not just hidden in the interface, so nobody can get around their role by changing something in the browser. This also applies to offline mode. Readings stored on a phone still need to be tied to a user and checked against that user's role when they sync.

Role-Based Access Overview			
What each role can see and do in the moisture data system			
	Collector Data collectors	Researcher Read-only analysis	Supervisor Manages staff & fields
VIEWING			
View moisture data (current & historical)	Assigned field	All fields	All fields
DATA			
Upload new moisture readings	Assigned field	—	All fields
Delete own readings (corrections)	✓	—	✓
Edit / delete any reading	—	—	✓
ADMINISTRATION			
Create & modify fields	—	—	✓
Assign users to fields	—	—	✓
Assign roles to new employees	—	—	✓
✓ Permitted — Not permitted Text = permitted, scoped to the fields shown			

Figure 3. Role-based access overview showing what each role can view and do.

Owen Gilbertson

After meeting as a group last Wednesday, have been brainstorming ideas to incorporate into our projects for things in the project requirements. It was very beneficial being able to hear Jason answer some questions on Wednesday to understand the project better.

Megan Chng

Took notes on what data that needs to be stored in the app and what analysis needs to be done in real time on the field and what are considered more long-term analysis that the researchers and data collectors will need. Organizing the data into tables and mapping out what real time analysis will be needed to implement. Reaching out to Dr.Rover to get in contact with Dr.Jones.

Cael Shroff

Created a document to store and summarize main points and takeaways from group meetings, meetings with client, and for future meetings with faculty advisor. Reviewed previous meetings video and transcript to create notes on the new document for passed sessions.

Group

Finished up the lightning talk slides to be prepared for Tuesday. Created questions to fill in the gaps of our understanding. Reviewed prototype with Jason to determine a more polished project.

Pending Issues

Carter Voelkel

None reported.

Jonah Samuels

None reported.

Connor Moroney

None reported.

Owen Gilbertson

None reported.

Megan Chng

Still no communication from Dr.Jones, will reach out to Dr.Rover to find a solution

Cael Shroff

None reported.

Individual Contributions

Name	Individual Contributions	Hours This Week	Cumulative Hours
Connor Moroney	Mapped out potential user roles/abilities	6	12
Jonah Samuels	Brainstorming solutions to problem, prototyping	6	12
Carter Voelkel	Team Documentation/Organization, and research	6	12
Megan Chng	Began organizing data storage and analyzation	6	11
Cael Shroff	Documentation of previous meetings and takeaways	6	12
Owen Gilbertson	Brainstorming project requirements	6	12

Comments and Extended Discussion

No additional comments at this time.

Plans for the Upcoming Week**Carter Voelkel**

Research more + determine IDE to try out

Jonah Samuels

Acquire SCIO device from Jason, polish prototype using feedback from last weeks meeting.

Connor Moroney

The main task for next week is researching how to make the app work offline. Since corn fields have unreliable cell reception, collectors need to be able to enter readings without a connection and not lose any data. This includes looking into how a web app can be cached on a phone so it still opens with no signal, and how readings can be stored on the device and synced to the server once a connection returns. It also means checking that the averaging and variance check from our prototype can run fully offline.

Another goal is getting the project started on Git using the framework we choose, so the team has a shared place to begin building. By the end of the week, the plan is to have a recommended offline approach to bring back to the team and share with Jason at our next bi-weekly meeting.

Owen Gilbertson

Communicate project requirements with team, and brainstorm other needs we may not be thinking of

Megan Chng

Have a generally detailed ER model of data, see an example of the Excel sheet, decide any other analysis

Cael Shroff

Transcribe group meeting and meeting with advisor. Document key plans for design and implementation.

Summary of Weekly Advisor Meeting

We were unable to contact Dr. Jones this week, so we did not have an advisor meeting. Our follow-up email also went unanswered, so we plan to stop by his office to meet with him in person. If that doesn't work, we will talk with Dr. Rover about the best way to reach him.